ELBO and Variational Autoecoder

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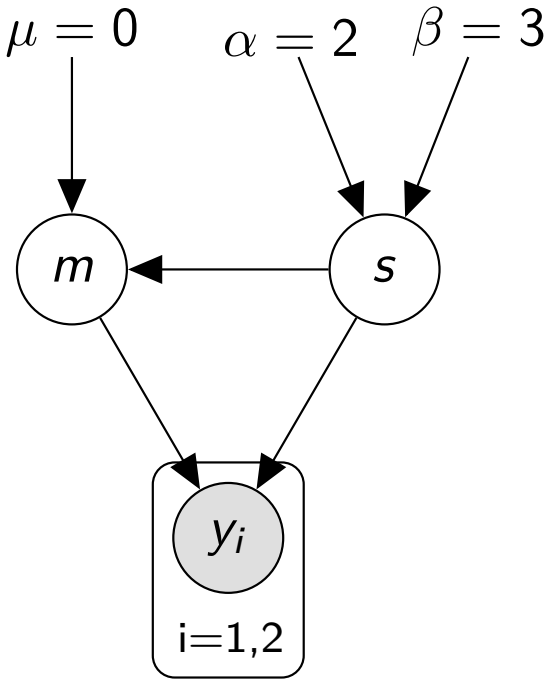
- what optimizes Variational Bayes?
- generative model of PCA? linear model $x = Az$,

Previous models



$$
\begin{aligned}
p(s) &= iG(\alpha_0, \beta_0) \\
p(m|s) &= \mathcal{N}(\mu, s) \\
p(y_i|m, s) &= \mathcal{N}(m, s)
\end{aligned}
$$

- Observations x_i are sampled from Gaussian with unknown mean and variance.
- We have some prior information about the mean and variance

Approximation via Variational Bayes

s

Divergence minimization

We seek best approximation of intractable distribution $p(x)$ in the chosen class of parametric functions, $q(x|\theta)$, such that

$$\theta^* = \arg\min_{\theta} D(p, q),$$

where $D(p, q)$ is a statistical divergence.

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Variational Bayes:

1. conditional independence $q(\theta_1, \theta_2) = q(\theta_1)q(\theta_2)$,
2. (reverse) Kullback-Leibler. $\mathrm{KL}(q||p) \neq \mathrm{KL}(p||q)$

Kullback Leibler divergence

Measure of divergence between two probability densities

$$KL(q||p) = E_q\left(\log\frac{q}{p}\right)$$

Not a metric!

$$E_q\left(\log\frac{q}{p}\right) \neq E_p\left(\log\frac{p}{q}\right)$$

also known as relative/free entropy

$$KL(q||p) = E_q\left(\log q\right) - E_q\left(\log p\right)$$

with properties:

1. $KL(q||p) \geq 0$,
2. $KL(q||p) = 0, \iff q = p$

Variational Bayes

Is a divergence minimization technique with

$$q^* = \arg\min_q KL(q||p) = \arg\min_q E_q\left(\log\frac{q}{p}\right)$$

$$q(m,s) = q(m|y_1,y_2)q(s|y_1,y_2).$$

which allows *free-form* optimization.

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which allows *free-form* optimization.
Result:

$$q(m|y_1,y_2) \propto \exp\left(E_{q(s)}\left[\log p(y_1,y_2,m,s)\right]\right)$$
$$q(s|y_1,y_2) \propto \exp\left(E_{q(m)}\left[\log p(y_1,y_2,m,s)\right]\right)$$

which is a set of implicit functions.

- Proportionality above allows to use $p(y_1,y_2,m,s)$ in place of $p(m,s|y_1,y_2)$
- Variational EM algorithm (E-E algorithm).

Toy: Variational Bayes

Factors:

$$q(m) = \mathcal{N}\left(m; \left(\frac{1}{\phi} + \frac{2}{\hat{s}}\right)^{-1} \left(\frac{\mu}{\phi} + \frac{y_1 + y_2}{\hat{s}}\right), \left(\frac{1}{\phi} + \frac{2}{\hat{s}}\right)^{-1}\right)$$

$$q(s) = i\mathcal{G}\left(\alpha_0 + 1, \mathrm{E}(m^2) - \mathrm{E}(m)(y_1 + y_2) + 0.5(y_1^2 + y_2^2) + \beta_0\right)$$

with

$$\hat{s} = \frac{\mathrm{E}(m^2) - \mathrm{E}(m)(y_1 + y_2) + 0.5(y_1^2 + y_2^2) + \beta_0}{\alpha_0 + 1},$$

$$\mathrm{E}(m) = \left(\frac{1}{\phi} + \frac{2}{\hat{s}}\right)^{-1} \left(\frac{\mu}{\phi} + \frac{y_1 + y_2}{\hat{s}}\right),$$

$$\mathrm{E}(m^2) = \mathrm{E}(m)^2 + \left(\frac{1}{\phi} + \frac{2}{\hat{s}}\right)^{-1},$$

which needs to be (Iterated).

Direct optimization of KL

Is a divergence minimization technique with

$$
\begin{aligned}
q^* &= \arg\min_q \mathrm{KL}(q||p) = \arg\min_q \mathrm{E}_q\left(\log\frac{q}{p}\right) \\
q(m,s) &= q(m|y_1,y_2)q(s|y_1,y_2). \\
q(m|y_1,y_2) &= \mathcal{N}(\mu_m,\sigma_m) \\
q(s|y_1,y_2) &= i\mathcal{G}(\alpha_s,\beta_s)
\end{aligned}
$$

Solving task:

$$
\mu_m^*,\sigma_m^*,\alpha_s^*,\beta_s^* = \arg\min_{\mu_m,\sigma_m,\alpha_s,\beta_s} \mathrm{E}_q\left(\log\frac{q}{p}\right),
$$

using general purpose black-box optimizer (SGD).

Loss function

$$
\begin{aligned}
\mathrm{KL}(q||p) &= \mathrm{E}_{q_m q_s}\left(\log \frac{q(m|x)q(s|x)}{p(m,s|x)}\right) \\
&= \mathrm{E}_{q_m q_s}\left(\log q(m|x) + \log q(s|x) - \log p(x,m,s) + \log p(x)\right), \\
&= -H(q(m|x)) - H(q(s|x)) - \mathrm{E}(\log p(x,m,s)) + \log p(x)
\end{aligned}
$$

Where

$$
\mathrm{E}_{q_m q_s}\left(\log p(x,m,s)\right) =
$$

$$
\mathrm{E}_{q_m q_s}\left[-(\alpha_0+2)\log s - \frac{1}{2}\frac{(m-y_1)^2}{s} - \frac{1}{2}\frac{(m-y_2)^2}{s} - \frac{1}{2}\frac{(m-\mu)^2}{\phi} - \frac{\beta_0}{s}\right] =
$$

$$
\begin{aligned}
&= -(\alpha_0+2)\mathrm{E}_{q_s}(\log s) - \mathrm{E}_{q_s}\left(\frac{\beta_0}{s}\right) \\
&\quad - \mathrm{E}_{q_s}\left(\frac{1}{2s}\right)\mathrm{E}_{q(m)}\left[(m^2 - 2my_1 + y_1^2 + m^2 - 2y_2 m + y_2^2\right] \\
&\quad - \frac{1}{2\phi}\mathrm{E}_{q(m)}\left[m^2 - 2m\mu + \mu^2\right]
\end{aligned}
$$

Loss function II

Using moments

$$\mathrm{E}_{q_m}(m) = \mu_m \qquad \mathrm{E}_{q_m}(m^2) = \mu_m^2 + \sigma_m^2,$$

$$\mathrm{E}_{q_s}(\frac{1}{s}) = \frac{\alpha_s}{\beta_s}, \qquad \mathrm{E}_{q_s}(\log s) = \log \beta - \psi(\alpha)$$

we obtain all components of KL:

$$\begin{aligned}
\mathrm{E}_{q_m q_s}\left(\log p(x, m, s)\right) = &- \left(\alpha_0 + 2\right)\left(\log \beta_s - \psi(\alpha_s)\right) - \frac{\beta_0 \alpha_s}{\beta_s} \\
&- \frac{\alpha_s}{\beta_s}\left[\mu_m^2 + \sigma_m^2 - \mu_m(y_1 + y_2) + \frac{y_1^2 + y_2^2}{2}\right] \\
&- \frac{1}{2\phi}\left[\mu_m^2 + \sigma_m^2 - 2\mu_m \mu + \mu^2\right] \\
H(q(m|x)) = &\frac{1}{2}\log(2\pi e \sigma_m^2) \\
H(q(s|x)) = &\alpha + \log\left(\beta \Gamma(\alpha)\right) - (\alpha + 1)\psi(\alpha)
\end{aligned}$$

KL and ELBO

Bayes rule for "data" x and "parameters" z:

$$p(z|x) = \frac{p(x,z)}{p(x)}, \quad p(x) = \int p(x|z)p(z)dz$$

Approximation

$$p(z|x) \approx q(z|x)$$

Divergence:

$$\begin{aligned}
\mathrm{KL}(q||p) &= \mathrm{E}_q\left(\log \frac{q(z|x)}{p(z|x)}\right) = \mathrm{E}_q\left(\log \frac{q(z|x)p(x)}{p(x,z)}\right) \\
&= \mathrm{E}_q\left(\log q(z|x) + \log p(x) - \log p(x,z)\right), \\
&= -H(q(z|x)) - \mathrm{E}_q(\log p(x,z)) + \log p(x)
\end{aligned}$$

The normalization $p(x)$ can either:

- neglected for minimization of the KL (constant).
- lower bounded (evidence lower bound):

$$\begin{aligned}
\log p(x) &= H(q(z|x)) + \mathrm{E}_q(\log p(x,z)) + \mathrm{KL}(q||p) \\
&\geq H(q(z|x)) + \mathrm{E}_q(\log p(x,z)) = \mathcal{L}(q(z|x))
\end{aligned}$$

KL minimization

Variational Bayes:

- Free-form: $q(m, s) \approx q(s)q(m)$ where forms of $q()$ are identified
- mean field Variational Bayes

ELBO:

- Direct: $q(m, s)$ is chosen by designer.
- conditionally independent $q(m)q(s)$ (same results)
- they can be **conditioned** $q(m|s)q(s)$
- often slower optimization (tuning SGD)
- allows **non-linear** transformations

Reparametrization trick

Variational Bayes requires knowledge of the moments $q_m = \mathcal{N}(\mu_m, \sigma_m)$

$$\mathrm{E}_{q_m}(m) = \mu_m \qquad\qquad \mathrm{E}_{q_m}(m^2) = \mu_m^2 + \sigma_m^2,$$

in general we may encounter moments, e.g. $\mathrm{E}(\log m^2)$.

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Approach: Monte Carlo

$$\mathrm{E}_{q(m)}(f(m)) \approx \frac{1}{N}\sum_{i=1}^{N} f(m^{(i)}), \;\; m^{(i)} \sim q(m)$$

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Approach: Monte Carlo

$$\mathrm{E}_{q(m)}(f(m)) \approx \tfrac{1}{N}\textstyle\sum_{i=1}^{N} f(m^{(i)}), \;\; m^{(i)} \sim q(m)$$

For $q(m) = \mathcal{N}(\mu_m, \sigma_m)$ we can approximate, $m^{(i)} = \mu_m + \sigma_m e^{(i)}$

$$\begin{aligned}\mathrm{E}_{q(m)}(f(m)) &\approx \tfrac{1}{N}\textstyle\sum_{i=1}^{N} f(m^{(i)}) = \tfrac{1}{N}\textstyle\sum_{i=1}^{N} f(\mu_m + \sigma_m e^{(i)}) \\ &= \mathrm{E}_{p(e)}(f(\mu_m + \sigma_m e))\end{aligned}$$

- Exact for large N (GD).
- **Unbiased** estimate for low N, even $N = 1$ (SGD).
- Variance reduction (decreasing learning rate, iterative averaging)

Density estimation

- We have: samples $X = \{x_1, \ldots x_n\}$
- We seek: probability density $p(x)$ from which they were generated
  - ideally such that we can generate artificial samples

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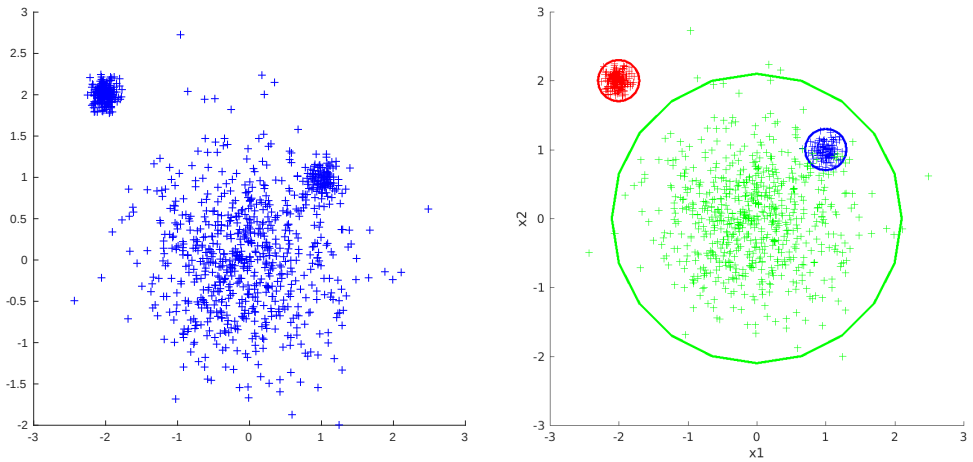
Examples (Mixture):

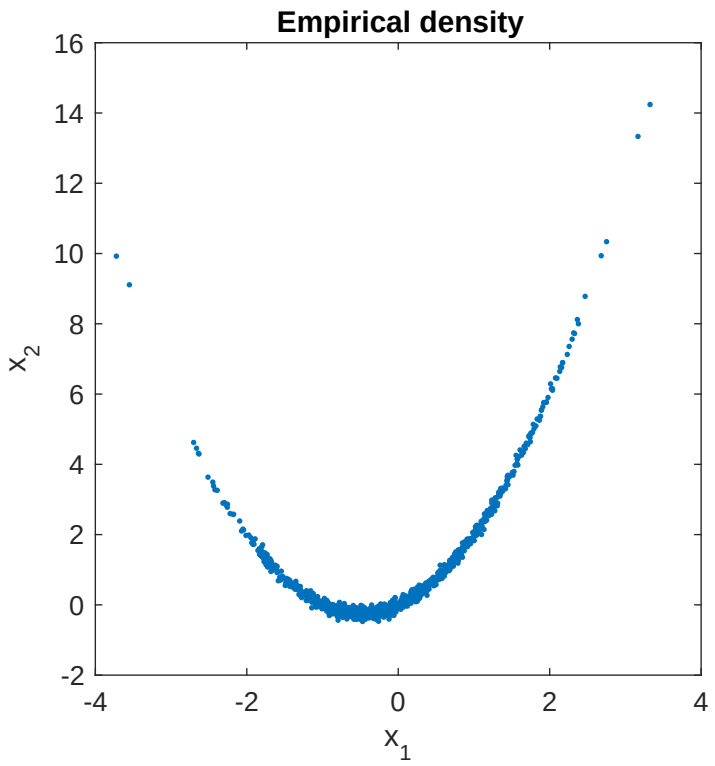


Density model is a mixture with parameters $\alpha_1 \dots, \alpha_K$, $\mu_1, \dots, \mu_K$ and $\Sigma_1, \dots, \Sigma_K$.

- maximum likelihood estimate by gradient descent
- EM algorithm (coordinate descent on negative ELBO)

Generative model

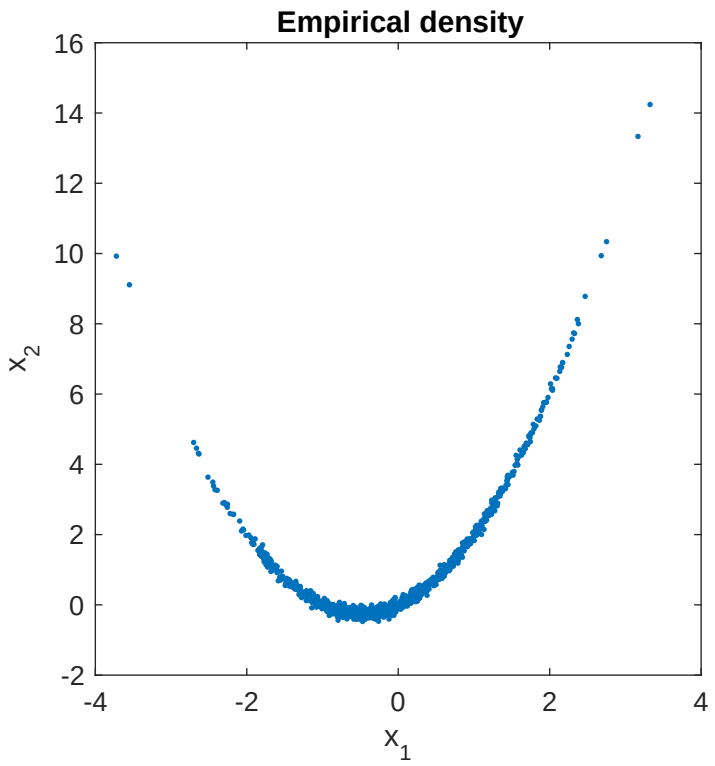
Mixture model?



Generative model

Mixture model?
True generative model:

$$z \sim \mathcal{N}(0,1),$$
$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} z \\ z + z^2 \end{bmatrix} + 0.1e$$



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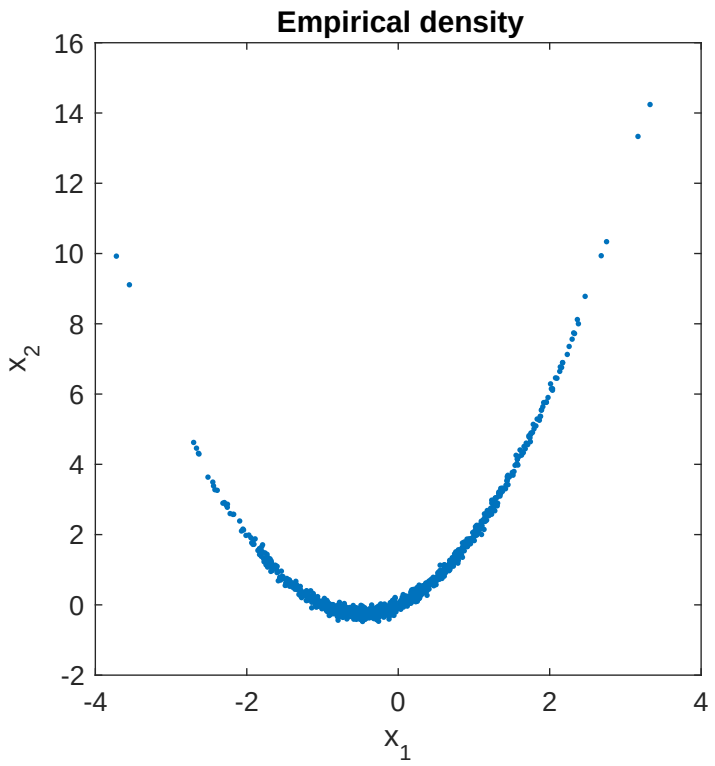
Class of models:

$$z \sim \mathcal{N}(0, I)$$
$$x \sim \mathcal{N}(f_\theta(z), \sigma I),$$

Marginal

$$p(x|\theta) = \int p(x|z,\theta)p(z)dz$$

Find θ. Maximum (marginal) likelihood.



Generative models

Flow models:

$$x = f_\theta(z), \quad p_z(z) = \mathcal{N}(0, I)$$

using transformation of coordinates

$$p(x) = p_z(f_\theta^{-1}(x)) \left| \frac{\partial f_\theta^{-1}(x)}{\partial x} \right|.$$

- only invertible transformations f
- equal dimension of z and x
- exact optimization of θ

Autoencoders:

$$x = f_\theta(z) + \sigma e, \quad p_z(z) = \mathcal{N}(0, I)$$
$$p(e) = \mathcal{N}(0, I)$$

with

$$p(x) = \int p(x|z)p(z)dz.$$

- z can have arbitrary dimension
- no need for inversion of f,
- no analytical solution of $p(x)$

Variational Autoencoder [Kingma, Welling, 2014]

Generative model

$$x = f_\theta(z) + \sigma e, \qquad p_z(z) = \mathcal{N}(0, I), \qquad p(e) = \mathcal{N}(0, I)$$

with

$$p(x) = \int p(x|z)p(z)dz \approx \mathcal{L}(q(z|x)).$$

With choices:

$$q(z|x) = \mathcal{N}(\mu_\psi(x), \mathrm{diag}(\sigma_\psi(x)^2)),$$

- $f_\theta(z)$ is a neural network
- $\mu_\psi(x), \sigma_\psi(x)$, is a neural network

Variational Autoencoder [Kingma, Welling, 2014]

Elaborate:

$$\begin{aligned} p(x|\theta,\psi) \approx & \mathrm{E}_{q(z|x)}\left(\log \frac{p(z,x)}{q(z|x)}\right) = \mathrm{E}_{q(z|x)}\left(\log \frac{p(x|z)p(z)}{q(z|x)}\right) \\ & = \mathrm{E}_{q(z|x)}\left(\log p(x|z)\right) + \mathrm{E}_{q(z|x)}\left(\log \frac{p(z)}{q(z|x)}\right) \\ & = \mathrm{E}_{q(z|x)}\left(\log p(x|z)\right) + \mathrm{KL}\left(q(z|x)||p(z)\right) \end{aligned}$$

Recall that $x = f_\theta(z) + \sigma e, p(x|z) = \mathcal{N}(f_\theta(z), I)$

$$\begin{aligned} \mathrm{E}_{q(z|x)}\left(\log p(x|z)\right) &= -\mathrm{E}_{q(z|x)}\left[(x - f(z))^2/\sigma^2\right] \\ \mathrm{KL}\left(q(z|x)||p(z)\right) &= \sum(-2\log(\sigma_\psi(x)) + \sigma_\psi(x)^2 + \mu_\psi(x)^2) \end{aligned}$$

yielding optimization (with reparametrization trick)

$$\theta^*, \psi^* = \arg\min_{\theta,\psi} \sum_{i\in\mathcal{I}} \left((x_i - f_\theta(\mu_\psi(x_i) + \sigma_\psi(x_i)e_i)^2\right) + \mathrm{KL}$$

which is known as autoencoder structure in NN. (for $\sigma_\psi = 0$, KL=0)

Linear model

We have analyzed such model

$$x = Az + e,$$

with analytical solution $p(x) = \mathcal{N}(0, AA^T + \sigma I)$. We can also compute

$$p(z|x) =$$

Linear model

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$$x = Az + e,$$

with analytical solution $p(x) = \mathcal{N}(0, AA^T + \sigma I)$. We can also compute

$$p(z|x) = \mathcal{N}((A^T A)^{-1} A^T x, (A^T A)^{-1})$$

In ELBO, need to choose approximation:

$$q(z|x) = \mathcal{N}(Bx, \mathrm{diag}(\beta^2)).$$

Then VAE is:

$$A^*, B^*, \beta^* = \arg\min_{A,B,\beta} \sum_{i \in \mathcal{I}} \Big[\sigma^{-1} \left|\left| x_i - A(Bx_i + \beta \circ e_i) \right|\right|^2 + \sum_j \left(-2\log(\beta_j) + \beta_j^2 + (\underline{B}_j x_i)^2 \right) \Big]$$

Explicit density of VAE (reduced dimensions)

Approximation of:

$$p(x) = \int p(x|z)p(z)dz \approx \mathcal{L}(q(z|x)).$$

- Simple approximation:

$$p(x|z = g(x)) = \int p(x|z)\delta(z - g(x))dz$$

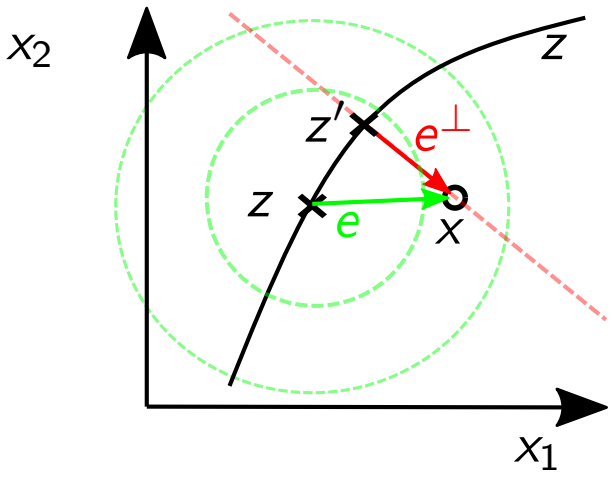
- No noise

$$p(x) \approx p_z(g(x)) \left| \frac{\partial g(x)}{\partial x} \right|$$

- Orthogonal approximation

$$x = f(z') + e'$$

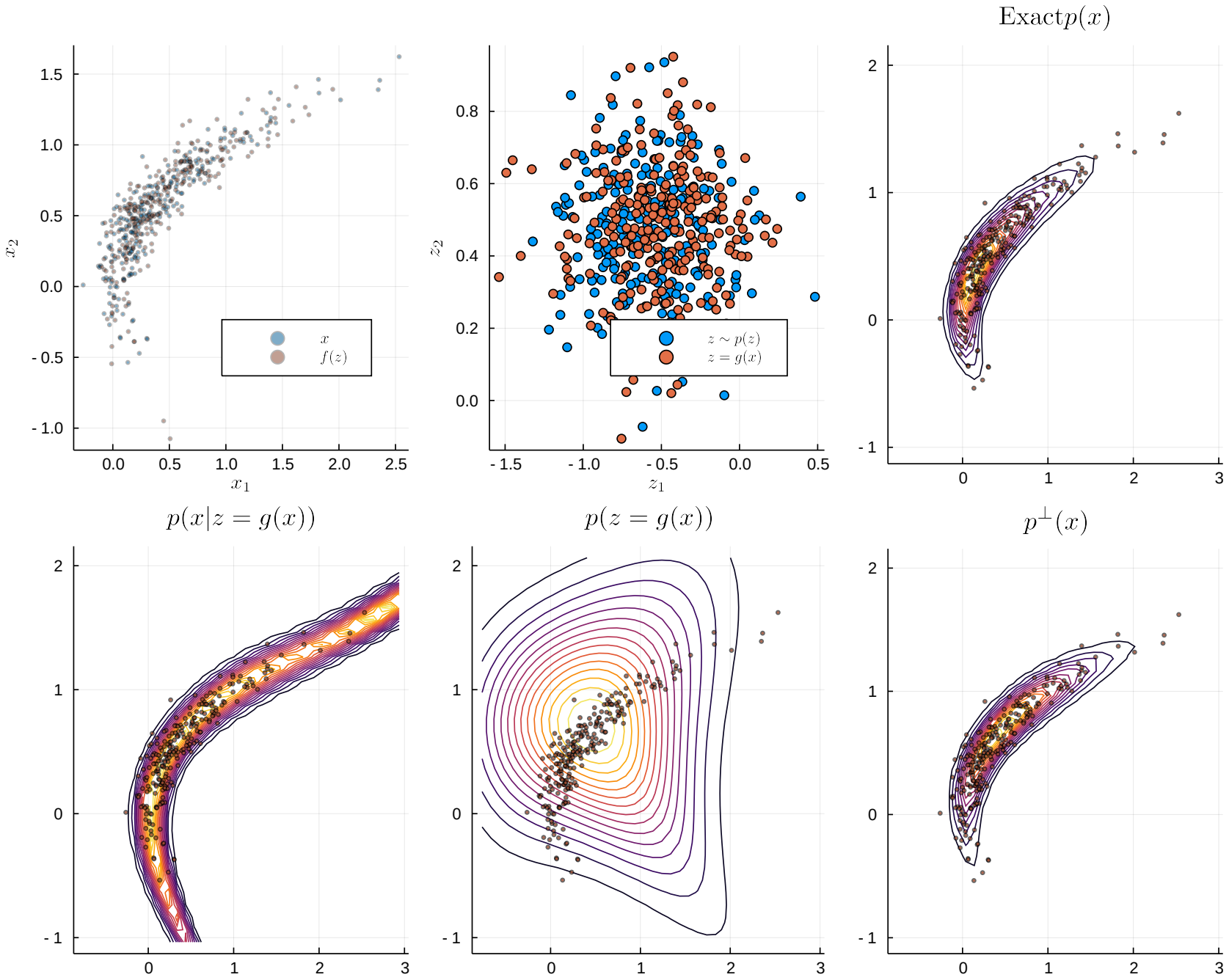
where $f(z')$ and e' are orthogonal.



Then

$$p(x) \approx \mathcal{N}(z'|0, \mathbf{I}) \left| \frac{\partial f^{-1}(z')}{\partial z'} \right| \mathcal{N}(x - f(z'), \sigma^2 \mathbf{I}).$$

Results



Variational Autoencoder

- Scales well with dimension (unlike GMM)
- Many extensions
  - two stage vae [Dai&Wipf,2019]: $x = f_z(z),\ z = f_w(w)$
  - First stage ($x = f_z(z)$) only reduces dimension $\dim(z) < \dim(x)$
  - Second stage ($z = f_w(w)$) models distribution $p(w) = \mathcal{N}(0, I)$
- Infinite-stage VAE is diffusion [Kingma, Salimans, Poole, Ho, 2021]
- Competes with GANs (without probabilistic density).

Diffusion as infinite-stage VAE

Two-stage VAE: $x = f_z(z)$, $z = f_w(w)$.

Diffusion as infinite-stage VAE

Two-stage VAE: $x = f_z(z)$, $z = f_w(w)$.
Generalise to T stages with *fixed* (non-learned) linear encoder:

$$x_t = \sqrt{1-\beta_t}\, x_{t-1} + \sqrt{\beta_t}\, \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I), \quad t = 1, \ldots, T$$

with $x_0 \equiv x$ (data) and noise schedule $0 < \beta_1 < \cdots < \beta_T < 1$.

Diffusion as infinite-stage VAE

Two-stage VAE: $x = f_z(z)$, $z = f_w(w)$.
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with $x_0 \equiv x$ (data) and noise schedule $0 < \beta_1 < \cdots < \beta_T < 1$.

- Each step adds a small Gaussian perturbation (forward / diffusion process).
- Marginal converges: $x_T \approx \mathcal{N}(0, I)$ for large T.
- Decoder *reverses* the diffusion: reconstruct x_0 from x_T.

Forward process

Closed-form marginal at step t (no need to iterate):

$$q(x_t|x_0) = \mathcal{N}\left(\sqrt{\bar{\alpha}_t}\, x_0,\ (1-\bar{\alpha}_t)I\right),$$

where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.

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where $\bar{\alpha}_t = \prod_{s=1}^{t}(1-\beta_s)$.
Reparametrised sample:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

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Reparametrised sample:

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$

- $\bar{\alpha}_t \to 0$ as $t \to T$: signal is destroyed, only noise remains.
- Posterior $q(x_{t-1}|x_t, x_0)$ is Gaussian (tractable):

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \mathcal{N}\left(\tilde{\mu}_t(x_t, x_0),\ \tilde{\beta}_t I\right), \\ \tilde{\mu}_t &= \frac{\sqrt{\bar{\alpha}_{t-1}}\,\beta_t}{1-\bar{\alpha}_t}\, x_0 + \frac{\sqrt{1-\beta_t}\,(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\, x_t, \\ \tilde{\beta}_t &= \frac{(1-\bar{\alpha}_{t-1})\beta_t}{1-\bar{\alpha}_t}. \end{aligned}$$

Reverse process

We cannot evaluate $q(x_{t-1}|x_t)$ without x_0, so we *learn* it:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \sigma_t^2 I\big)\,.$$

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Generative (sampling) procedure:

1. Draw $x_T \sim \mathcal{N}(0, I)$.
2. For $t = T, T-1, \ldots, 1$: sample $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$.
3. Return x_0.

Reverse process

We cannot evaluate $q(x_{t-1}|x_t)$ without x_0, so we *learn* it:

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}\big(\mu_\theta(x_t, t),\ \sigma_t^2 I\big)\,.$$

Generative (sampling) procedure:

1. Draw $x_T \sim \mathcal{N}(0, I)$.
2. For $t = T, T-1, \ldots, 1$: sample $x_{t-1} \sim p_\theta(x_{t-1}|x_t)$.
3. Return x_0.

Analogy with VAE:

- **Encoder**: fixed forward chain $q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$.
- **Decoder**: learned reverse chain $p_\theta(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p_\theta(x_{t-1}|x_t)$.

ELBO for diffusion

By the same ELBO derivation as VAE:

$$
\begin{aligned}
\log p_\theta(x_0) &\geq \mathcal{L}(\theta) \\
&= -\mathrm{KL}(q(x_T|x_0) \,\|\, p(x_T)) \\
&\quad - \sum_{t=2}^{T} \mathrm{E}_q[\mathrm{KL}(q(x_{t-1}|x_t, x_0) \,\|\, p_\theta(x_{t-1}|x_t))] \\
&\quad + \mathrm{E}_q[\log p_\theta(x_0|x_1)] \,.
\end{aligned}
$$

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- First term: ≈ 0 (prior matching, fixed schedule).
- **Denoising terms**: each KL between two Gaussians $\Rightarrow$ closed form.
- Reconstruction term: as in VAE.

Training: noise prediction

Both $q(x_{t-1}|x_t, x_0)$ and $p_\theta(x_{t-1}|x_t)$ are Gaussian $\Rightarrow$

$$\mathrm{KL}(q \,\|\, p_\theta) \propto \|\tilde{\mu}_t(x_t, x_0) - \mu_\theta(x_t, t)\|^2 .$$

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Substitute $x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}\left(x_t - \sqrt{1-\bar{\alpha}_t}\,\epsilon\right)$ and reparametrise μ_θ:

$$\mu_\theta(x_t, t) = \frac{1}{\sqrt{1-\beta_t}}\left(x_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(x_t, t)\right),$$

where $\epsilon_\theta(\cdot, t)$ is a neural network predicting the noise.

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Simplified training objective [Ho et al., 2020]:

$$\theta^* = \arg\min_\theta \; \mathrm{E}_{t,\,x_0,\,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\,x_0 + \sqrt{1-\bar{\alpha}_t}\,\epsilon,\; t\right)\right\|^2\right].$$

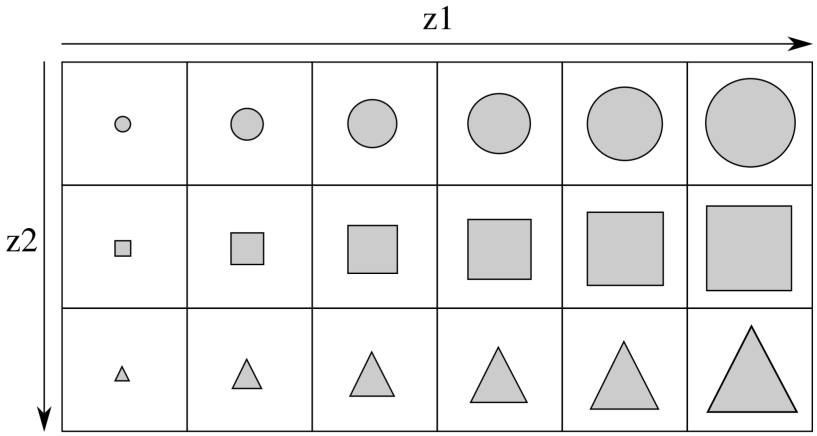
- Uniform $t \sim \mathcal{U}\{1, \ldots, T\}$, $\epsilon \sim \mathcal{N}(0, I)$, x_0 from data.
- Same MSE structure as VAE reconstruction loss.

Disentanglement

Original problem of linear source separation is rotation

$$X = AZ = (AT)(T^{-1}Z),$$

Sometimes we want the sources to have a meaning (Bart-Lisa).



Typically we assume that we have a partial information (observation) u such that

$$u = h(z)$$

and we want to learn the state variables that correspond to the meaning.

[Mita, Filippone, Michi, 2020] Learning Optimal Conditional Priors For Disentangled Representations.

Homework

Matrix factorization		points
Variational Autoencoder for toy probelm		8
+ sensitivity to dimension of z		+2